POWER BI REPORT GUIDE

Bank churn: who leaves, and where we can stop it

How the Bank Churn report was built, what each page shows, how to use the dashboards, and how to maintain and extend the project.



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Report: Bank Churn.pbip (15 pages: 11 story pages, 3 dashboards, 1 drill-through)
Data: Bank_Churn.xlsx, 10,307 customer records, 5,000 of them reliable
Tools: Power BI Desktop (PBIP format), Power BI modeling MCP, powerbi-report-author CLI, Node.js
Version: October 2026

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> **Who this guide is for.** Managers and analysts who will read or present the report (sections 1, 2, 8, 9, 10), and analysts who will maintain or extend it (sections 3 to 7 and 11 to 14).

1 · EXECUTIVE SUMMARY

23% of reliable customers have left, and behaviour, not demographics, explains why

23.1%	90.1M	62.3%	≈172
churn rate (1,155 of 5,000)	balances held by churners (24.2%)	of churners were inactive members	leavers two actions could prevent

Key findings

1. **Half the file is unreliable.** Rows 5,001 to 10,307 store their yes/no flags as text, churn at a flat 51% in every group and contradict their own band keys in 98% of cases. All findings use the 5,000 reliable rows.
2. **Inactivity nearly doubles churn:** 29.9% for inactive members against 16.8% for active ones. Inactive members are 48% of customers but 62% of churners.
3. **A third product adds risk instead of loyalty:** customers with 3 or more products churn at 40.8%, twice the 21% of those with 1 or 2.
4. **The two effects stack:** inactive members with 3+ products churn at 45.2%, three times the 14.5% of active members with 1 or 2 products.
5. **High-balance inactive customers are the costliest leavers:** 624 customers holding 77.6M, of whom 261 (41.8%) have left, taking 32.6M.
6. **Age matters a little:** customers over 45 churn at 25 to 26%, about 6 points above younger ones. Tenure does not protect at any length.
7. **Demographics do not separate leavers from stayers:** country, gender, credit score and salary all sit within about 2 points of 23%.

Recommended actions

Action	What to do	Target	Leavers avoided
1 Reactivate	Win back the 2,410 inactive members, starting with the 624 high-balance inactive.	Inactive churn from 29.9% to 25%	≈118
2 Review bundles	Find out why 3+ product holders leave; stop pushing a third product.	3+ churn from 40.8% to 30%	≈54
3 Fix the data	Trace where the 5,307 suspect rows came from before the next review.	0 suspect records	n/a

Together, actions 1 and 2 would cut churn from 23.1% to about 19.7%. Inactive 3+ product holders are in both groups, so treat the combined figure as an upper estimate.

A retail bank in three markets wants to know who leaves, why, and what to do

The bank operates in France, Germany and Spain. The data is a single snapshot of customers, with no transaction or churn dates and no stated currency. The audience is bank executives and retention leads, and the report accompanies the executive deck *Bank Churn Who Leaves and Why.pptx*.

Questions the report answers

Theme	Questions	Where answered
How big is the problem?	Overall churn and balances lost; whether the data can be trusted	Pages 2, 3, D1
Who leaves?	Effect of activity, product count, age, tenure, balance segment	Pages 4 to 8, D2
What does not matter?	Country, gender, credit score, salary	Page 9
What is at stake?	Which segment lost the most balance; how concentrated churn is	Page 8, D3
What should we do?	Actions, targets and the churn rate if they succeed	Page 10

Management beliefs tested

Belief	Verdict	Evidence
Churn is random across the base	FALSE	Inactive 30% vs active 17%
More products means stickier customers	PARTLY TRUE	1-2 products 21%; 3+ products 41%
High-balance customers are safe	PARTLY TRUE	High-balance active 15%, high-balance inactive 42%
Country and gender drive churn	FALSE	Every country and both genders about 23%
Older customers are the most loyal	FALSE	Over 45: 25-26%; under 45: about 19%

3 · THE SOURCE DATA

One Excel workbook with a customer table and six lookup sheets

`Bank_Churn.xlsx` has seven sheets. **FactCustomer** holds one row per customer (10,307 rows, 17 columns); the other six are small lookup tables.

Sheet	Rows	Columns
FactCustomer	10,307	CustomerID, Surname, CreditScore, GeographyID, GenderID, Age, Tenure_Years, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, HasChurned, AgeBandID, TenureBandID, ProductBandID, SegmentID
DimGeography	3	GeographyID, Country (France, Germany, Spain)
DimGender	2	GenderID, Gender
DimAgeBand	4	AgeBandID, Band (<30, 30-45, 46-60, 60+)
DimTenureBand	4	TenureBandID, Band (<2, 2-5, 6-9, 10+)
DimProductBand	3	ProductBandID, Band (1, 2, 3+)
DimSegment	4	SegmentID, SegmentName (High/Low Balance × Active/Inactive)

Field notes

- **HasCrCard, IsActiveMember, HasChurned** mix real true/false values with the text "TRUE"/"FALSE". This turned out to be the key to the data-quality finding (section 4).
- **Balance** is blank for 100 customers; currency is not stated anywhere.
- **AgeBandID, TenureBandID, ProductBandID, SegmentID** are supplied, but they contradict the raw Age, Tenure and Products values in much of the file, so the model rebuilds them.
- There are **no dates**. A join year is derived as snapshot year (2025) minus tenure.

4 · DATA QUALITY

The file splits cleanly at row 5,000 into a reliable block and a suspect block

Two earlier documents disagreed. The build guides used all 10,307 rows (37.6% churn); the executive deck used only the first 5,000 (23.1%). Profiling the workbook settled it: every quality signal changes exactly at row 5,000.

Signal	Rows 1-5,000	Rows 5,001-10,307
Churn rate	23.1%	51.3%
Churn by country, products, activity, age	Varies (activity 17% vs 30%; 3 products 42%)	Flat at 49-54% in every group
Band keys contradicting raw values	0%	98%
How yes/no flags are stored	Real true/false	Text
Blank or zero balances	226	None
Credit scores above 850	None	479

> **Decision.** Load all 10,307 rows, flag each with `IsReliable`, and filter every analysis page to the 5,000 reliable rows. The suspect block stays in the model so pages 3 and 11 can show it.

How the flag is computed

In Power Query, before the flags are normalised: a row is reliable when all three yes/no columns are real booleans. This matches exactly customer IDs 1 to 5,000.

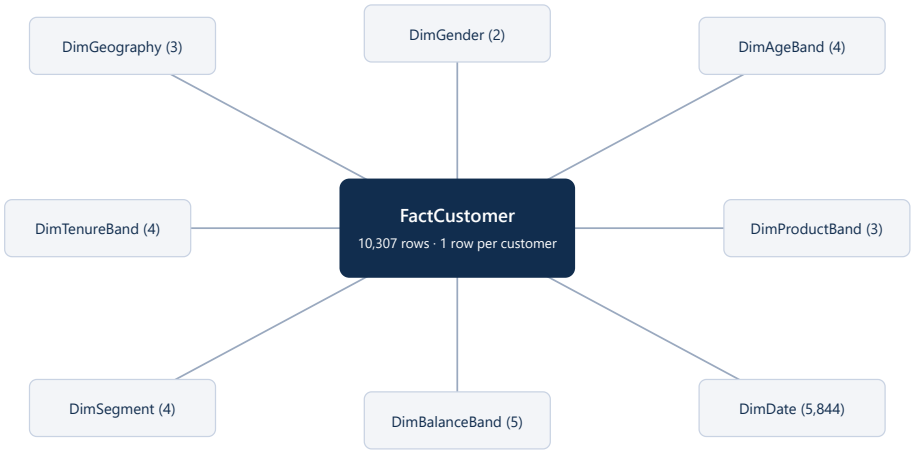
```
IsReliable = Value.Is([HasCrCard], type logical)
         and Value.Is([IsActiveMember], type logical)
         and Value.Is([HasChurned], type logical)
```

Other issues and how they are handled

Issue	Size	Handling
Flags stored as mixed text and boolean	3 columns	Normalised to true/false in Power Query
Blank balance	100 rows	Set to 0; `BalanceMissing` flag kept
Band and segment keys inconsistent with raw values	Most suspect rows	All four keys rebuilt from raw fields
Ages 80 or over	470 reliable, 1,349 suspect	Kept; `AgeImplausible` flag
Tenure longer than age minus 18	570 reliable, 403 suspect	Kept; `TenureExceedsAge` flag
Surname spelling variants and stray spaces	Many	Trimmed and proper-cased; hidden
No dates; no currency	All rows	Join year derived from tenure; balances shown unitless

5 · THE DATA MODEL

A star schema: one customer fact table and eight dimensions



All relationships are many-to-one from FactCustomer to the dimension, single direction, active. Dimensions never relate to each other, so there is one filter path to the fact table.

Table	Grain	Key	Source
FactCustomer	One customer	CustomerID	FactCustomer sheet, heavily transformed
DimGeography	One country	GeographyID	Sheet
DimGender	One gender	GenderID	Sheet
DimAgeBand	One age band	AgeBandID	Sheet; Band renamed Age Band
DimTenureBand	One tenure band	TenureBandID	Sheet; Band renamed Tenure Band
DimProductBand	One product band	ProductBandID	Sheet; Band renamed Product Band
DimSegment	One segment	SegmentID	Sheet
DimBalanceBand	One balance band	BalanceBandID	Entered in Power Query: 0, >0-50k, >50-100k, >100-150k, >150k
DimDate	One day, 2010-2025	Date	Generated in Power Query; marked as the date table

Model settings

- **Parameters:** `SourceFile` (full path to the workbook) and `SnapshotYear` (2025). Change these in Transform data → Manage parameters.
- **Sort order:** every band label sorts by its ID; Month sorts by Month No; CreditScoreBand sorts by CreditScoreBandSort.
- **Hidden:** all ID and key columns, Surname, CustomerID, JoinDate, raw Balance and EstimatedSalary (used through measures).
- **Hierarchy:** DimDate Join Date Hierarchy (Year › Quarter › Month). Join dates are always 1 January, so Year is the useful level.
- **Implicit measures discouraged:** numeric columns are not summed by accident; all numbers come from explicit measures.

6 · POWER QUERY

FactCustomer is cleaned, flagged and re-keyed in a single query

#	Step	What it does
1	Source, Sheet, Promoted	Read the FactCustomer sheet from `SourceFile` and promote headers
2	RemovedKeys	Drop the four supplied band/segment keys (they are wrong)
3	Reliable	Add `IsReliable` from the flag storage type (section 4)
4	Flags	Normalise HasCrCard, IsActiveMember, HasChurned: `Text.Upper(Text.From(_)) = "TRUE"`
5	Typed, CleanSurname	Set data types (fixed decimal for money); trim and proper-case Surname
6	BalanceMissing, BalanceFilled	Flag blank balances, then replace them with 0
7	AgeBand … BalanceBand	Rebuild AgeBandID (≤29, ≤45, ≤60, 61+), TenureBandID (≤1, ≤5, ≤9, 10+), ProductBandID (1, 2, 3+) and BalanceBandID
8	Segment	High balance = 100,000 or more, crossed with IsActiveMember → SegmentID 1-4
9	HighRisk	`HighRiskFlag` = 3+ products or balance over 150,000
10	CreditBand, CreditLabel	CreditScoreBand (<500 … 800+) with a sort column
11	JoinDate	1 January of (SnapshotYear − Tenure_Years)
12	AgeFlag, TenureFlag	`AgeImplausible` (80+) and `TenureExceedsAge` (tenure > age − 18)
13	Grouped, Expanded	Salary quintiles ranked separately inside the reliable and suspect blocks
14	Final, Sorted	Re-apply all types and sort by CustomerID
15	RecordBlock, Activity	Readable labels: "Rows 1-5,000 (reliable)" / "Rows 5,001-10,307 (suspect)"; "Active" / "Inactive"

Segment rule

```
SegmentID = if [Balance] >= 100000
            then (if [IsActiveMember] then 1 else 2)   // High Balance Active / Inactive
            else (if [IsActiveMember] then 3 else 4)   // Low Balance Active / Inactive
```

> The executive deck counted 632 high-balance inactive customers (265 leavers); the model counts 624 (261). The balance total is identical (77.6M), which suggests the deck's analysis treated 8 blank-balance inactive customers as high balance. The model's figure is the correct one.

42 explicit measures on FactCustomer, grouped into seven folders

All measures live on FactCustomer in display folders. They use only simple DAX (COUNTROWS, SUM, AVERAGE, CALCULATE, DIVIDE), and every one has a description in the model.

01 Core

Measure	DAX	Format
Customers	`COUNTROWS ( FactCustomer )`	#,0
Churned Customers	`CALCULATE ( [Customers], FactCustomer[HasChurned] = TRUE () )`	#,0
Retained Customers	`CALCULATE ( [Customers], FactCustomer[HasChurned] = FALSE () )`	#,0
Churn Rate	`DIVIDE ( [Churned Customers], [Customers] )`	0.0%
Retention Rate	`DIVIDE ( [Retained Customers], [Customers] )`	0.0%

02 Balance

Measure	DAX	Format
Total Balance	`SUM ( FactCustomer[Balance] )`	#,0 (shown in M on visuals)
Balance Lost	`CALCULATE ( [Total Balance], FactCustomer[HasChurned] = TRUE () )`	#,0
Balance Retained	`CALCULATE ( [Total Balance], FactCustomer[HasChurned] = FALSE () )`	#,0
Balance Lost %	`DIVIDE ( [Balance Lost], [Total Balance] )`	0.0%
Avg Balance / Churned / Retained	`AVERAGE ( FactCustomer[Balance] )`, then filtered by HasChurned	#,0

03 Behaviour

Measure	DAX	Format
Avg Products	`AVERAGE ( FactCustomer[NumOfProducts] )`	0.00
Active / Inactive Customers	`CALCULATE ( [Customers], FactCustomer[IsActiveMember] = TRUE () )` (FALSE for inactive)	#,0
Active Share	`DIVIDE ( [Active Customers], [Customers] )`	0.0%
Inactive Share of Customers	`DIVIDE ( [Inactive Customers], [Customers] )`	0.0%
Churn Rate Active / Inactive	`CALCULATE ( [Churn Rate], FactCustomer[IsActiveMember] = TRUE () )` (FALSE for inactive)	0.0%
Inactive Share of Churn	`DIVIDE ( CALCULATE ( [Churned Customers], FactCustomer[IsActiveMember] = FALSE () ), [Churned Customers] )`	0.0%

Card Holders	`CALCULATE ( [Customers], FactCustomer[HasCrCard] = TRUE () )`	#,0

04 Concentration

Measure	DAX	Format
Customers / Churned 3+ Products	`CALCULATE ( [Customers], FactCustomer[NumOfProducts] >= 3 )`	#,0
Churn Rate 3+ Products	`DIVIDE ( [Churned 3+ Products], [Customers 3+ Products] )`	0.0%
3+ Products Share of Customers / of Churn	`DIVIDE ( [Customers 3+ Products], [Customers] )` and the churn equivalent	0.0%
Customers / Churned High Risk	`CALCULATE ( [Customers], FactCustomer[HighRiskFlag] = TRUE () )`	#,0
High Risk Share of Customers / of Churn	`DIVIDE ( [Customers High Risk], [Customers] )` and the churn equivalent	0.0%
Overall Churn Rate	`CALCULATE ( [Churn Rate], ALLEXCEPT ( FactCustomer, FactCustomer[IsReliable] ) )`	0.0%
Churn vs Overall (pts)	`[Churn Rate] - [Overall Churn Rate]`	+0.0%;-0.0%

Overall Churn Rate keeps the IsReliable filter while removing everything else, so it stays a fair benchmark on the reliable base.

05 Scenarios

```
Leavers Prevented Reactivate =
    VAR _target = 0.25
    VAR _inactive = CALCULATE ( [Customers], FactCustomer[IsActiveMember] = FALSE () )
    RETURN MAX ( 0, _inactive * ( [Churn Rate Inactive] - _target ) )

Leavers Prevented Bundles =
    MAX ( 0, [Customers 3+ Products] * ( [Churn Rate 3+ Products] - 0.30 ) )

Leavers Prevented        = [Leavers Prevented Reactivate] + [Leavers Prevented Bundles]
Churn Rate After Actions = DIVIDE ( [Churned Customers] - [Leavers Prevented], [Customers] )
```

06 Data Quality and 07 Formatting

Measure	Purpose
Suspect Records	Count of rows where IsReliable = FALSE
Balance Missing / Age Implausible / Tenure Exceeds Age Records	Counts of each data-quality flag (page 11)
Churn Heat Color, Churn Heat Font Color (hidden)	Three-tier cell colours from the first build. The heat maps now use a gradient, so these two are no longer used and can be deleted.

8 · REPORT WALKTHROUGH

Story pages: eleven pages that follow the executive deck

Each story page makes one point. The title states the finding as a full sentence, the context bars are grey-blue, and only the bar or cell that proves the point is coloured, in orange-red for a problem or blue otherwise. Every page except 3 and 11 is filtered to the 5,000 reliable customers.

1 · Cover



Title, scope line and a **Start the story** button that jumps to page 2.

2 · Beliefs scorecard

Churn is 23%, and four of five management beliefs fail or only half hold

churn rate (1,155 of 5,000)	balance held by churned customers	of churners were inactive
23.1%	90.1M	62.3%

FALSE	Churn is random across the base	Inactive members churn at 30% vs 17% for active members
PARTLY TRUE	More products means stickier customers	1-2 products churn at 21%; 3+ products churn at 41%
PARTLY TRUE	High-balance customers are safe	High-balance active 15%, but high-balance inactive 42%
FALSE	Country and gender drive churn	Every country and both genders churn at about 23%
FALSE	Older customers are the most loyal	Over 45: 25-26%; under 45: about 19%

Source: Bank_Churn.xlsx, reliable base of 5,000 customers · 2

Dashboards

Three headline KPIs (churn rate, balances held by churners, share of churners who were inactive) and the five management beliefs, each with a verdict pill and the evidence. Read top to bottom; the verdict pill colour is always backed by the word.

3 · Data reliability

Half the file is unreliable: 5,307 records churn at 51% in every group

Churn rate by record block

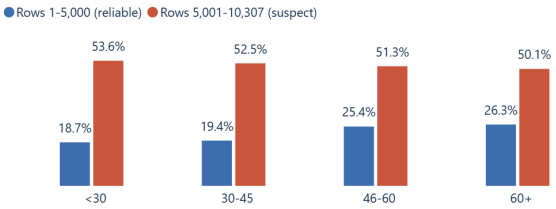


Band and segment keys contradict Age, Tenure and Products in 98% of the later rows (0% in the first 5,000). The later rows also have no zero or blank balances. All findings use the first 5,000 rows.

Churn rate by product band and record block (darker = higher)

Product Band	Rows 1-5,000 (reliable)	Rows 5,001-10,307 (suspect)
1	21.6%	50.8%
2	20.6%	51.1%
3+	40.8%	51.5%

Churn rate by age band: reliable rows vary, suspect rows stay flat



Source: Bank_Churn.xlsx, all 10,307 records · 3

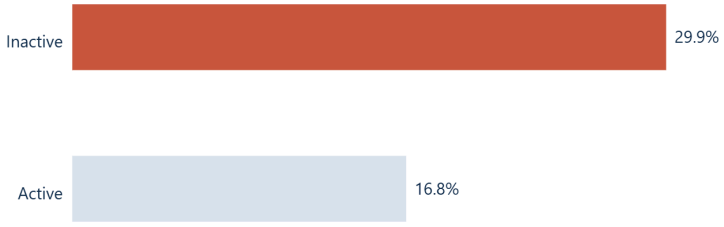
Dashboards

The proof that half the file is suspect. Left: churn by record block (23.1% vs 51.3%) and the key evidence. Right: a heat map by product band and record block, and churn by age band for both blocks. The reliable rows vary; the suspect rows sit at about 51% everywhere. **This page is not filtered.**

4 · Inactivity

Inactive members churn at 30%, nearly double the 17% of active members

Churn rate by membership activity



of customers are inactive
48.2%

of churners are inactive
62.3%

Inactive members are 48% of customers but 62% of churners.
Reactivating them is the biggest single lever (page 10).

Source: Bank_Churn.xlsx, reliable base of 5,000 customers · 4

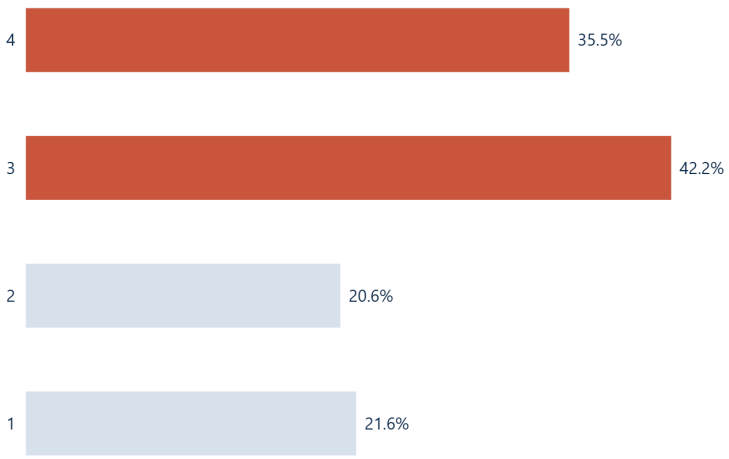
Dashboards

Inactive members churn at 29.9% against 16.8%. The cards show they are 48.2% of customers but 62.3% of churners, the over-representation that makes reactivation the biggest lever.

5 · Product count

Customers with 3 or more products churn at 41%, twice the rate of those with 1-2

Churn rate by number of products held



of customers hold 3+ products
10.1%

of churners hold 3+ products
17.7%

3+ product holders are 10% of customers but 18% of churners.
A third product adds risk instead of loyalty (page 10, action 2).

Source: Bank_Churn.xlsx, reliable base of 5,000 customers · 5

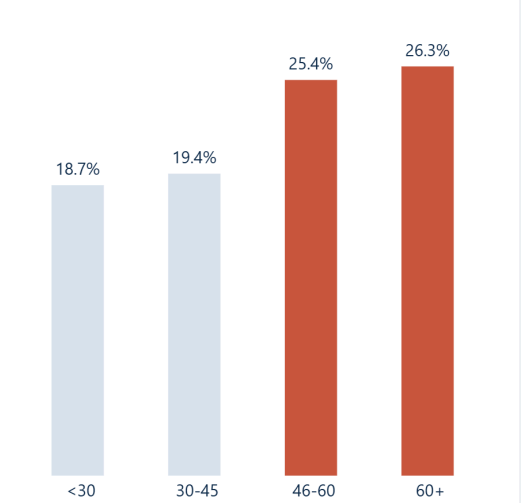
Dashboards

Churn by number of products: 3 products 42.2% and 4 products 35.5% (highlighted) against about 21% for 1 or 2. 3+ product holders are 10.1% of customers but 17.7% of churners.

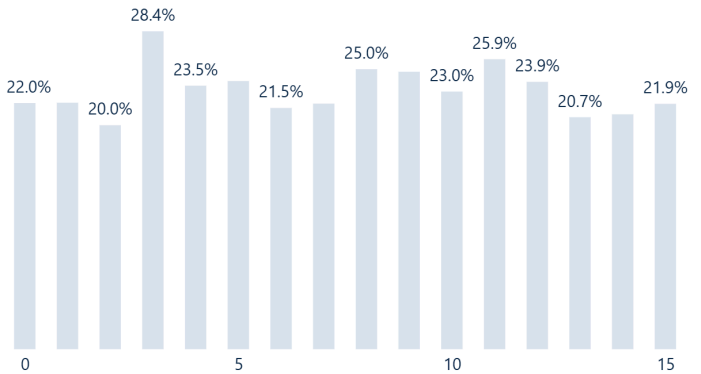
6 · Age and tenure

Customers over 45 churn at 25-26%, about 6 points above younger customers

Churn rate by age band



Churn rate by years with the bank



Tenure does not protect: churn is 20-28% at every tenure year from 0 to 15.

Source: Bank_Churn.xlsx, reliable base of 5,000 customers · 6

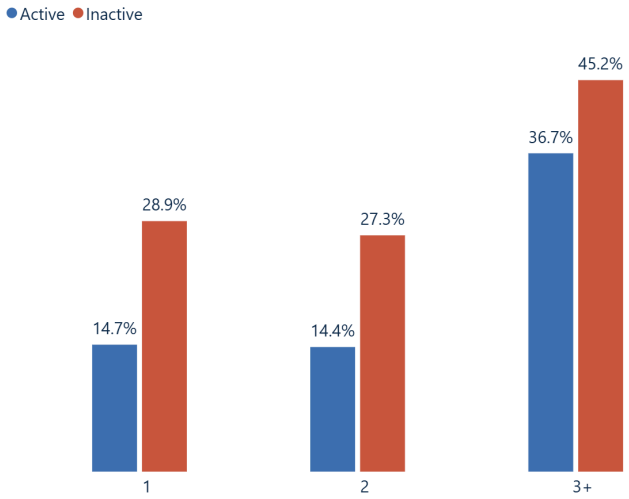
Dashboards

Churn rises from 18.7% under 30 to 26.3% over 60. Tenure gives no protection: every year from 0 to 15 sits between 20% and 28%.

7 · Activity × products

Inactive members with 3+ products churn at 45%, three times the rate of active members with 1-2

Churn rate by product band, active vs inactive



Same numbers as a heat map

Product Band	Active	Inactive
1	14.7%	28.9%
2	14.4%	27.3%
3+	36.7%	45.2%

How to read it
Darker cells mean higher churn.
Active customers with 1-2 products churn at 14.5%, the lowest-risk group.
Inactivity adds 13-14 points at every product count.

Source: Bank_Churn.xlsx, reliable base of 5,000 customers · 7

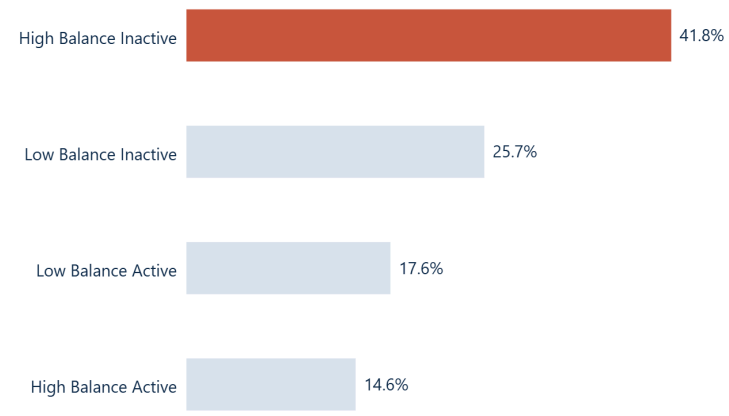
Dashboards

The two drivers combined. Inactive members churn more at every product band, peaking at 45.2% for inactive 3+ product holders. The heat map repeats the numbers; darker cells mean higher churn.

8 · Segments

High-balance inactive customers churn at 42%, nearly three times high-balance active ones

Churn rate by balance and activity segment



high-balance inactive customers
624

balances they hold
77.6M

of them have left
261

Reactivating these customers is the highest-value retention move.

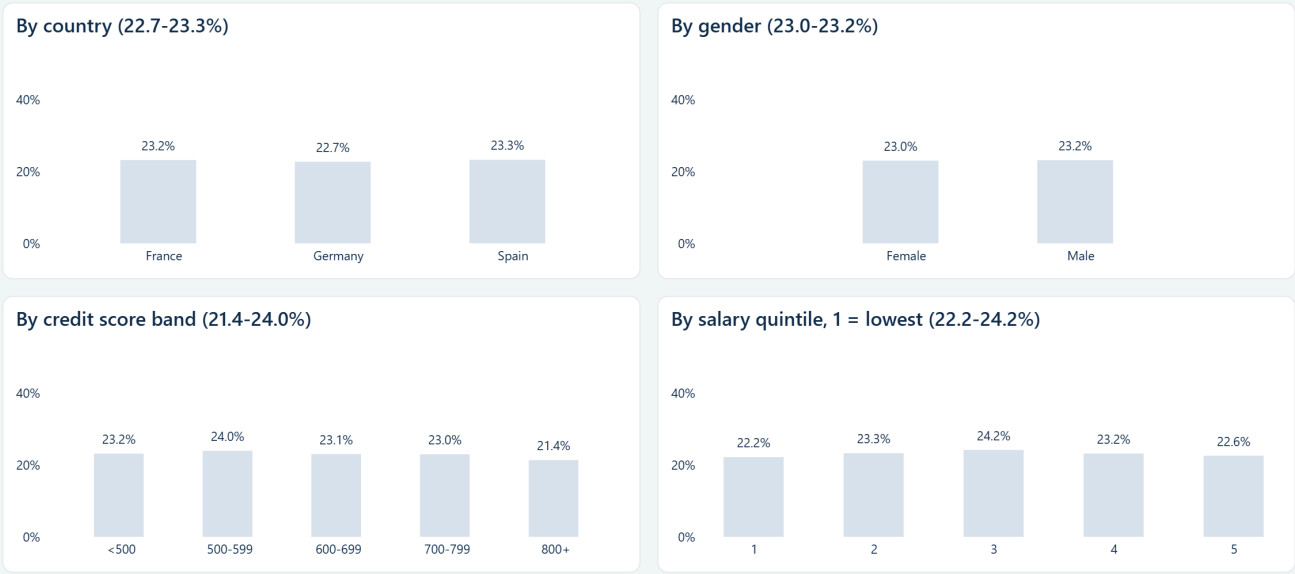
Source: Bank_Churn.xlsx, reliable base of 5,000; high balance = 100,000 or more · 8

Dashboards

High-balance inactive customers churn at 41.8%, nearly three times high-balance active ones (14.6%). The cards on the right are filtered to that segment: 624 customers, 77.6M in balances, 261 leavers.

9 · Demographics

Country, gender, credit score and salary do not separate leavers from stayers



Retention budgets should follow behaviour, not demographics. Source: reliable base of 5,000 · 9

Dashboards

Four small charts on the same 0-50% axis: country, gender, credit score band and salary quintile. All are flat around 23%, so retention budgets should follow behaviour, not demographics.

10 · Actions

Three actions can prevent about 170 departures: reactivate, review bundles, fix the data

1

Reactivate

Win back the 2,410 inactive members, starting with the 624 high-balance inactive.

Target: inactive churn from 29.9% to 25% (about 118 fewer leavers)

2

Review bundles

Audit why 3+ product holders leave and stop pushing a third product.

Target: 3+ churn from 41% to 30% (about 54 fewer leavers)

3

Fix the data

Find the source of the 5,307 flat-churn rows; band keys are already rebuilt in Power Query.

Target: 0 suspect records before the next review

churn today

23.1%

leavers prevented if both targets are met

172

churn after actions (upper estimate)

19.7%

The scenario adds both actions; inactive 3+ product holders are in both groups, so treat it as an upper estimate · 10

Dashboards

The three recommended actions with their targets, and the scenario KPIs: churn today (23.1%), leavers prevented if both targets are met (172) and churn after actions (19.7%, an upper estimate).

11 · Data issues

Assumptions and data issues behind these numbers

- No date field: no growth, trend or seasonality analysis is possible; join year is assumed as 2025 minus tenure.
- Currency is not stated; balances are unitless.
- Rows 5,001-10,307 store flags as text, churn at a flat 51% and contradict their band keys; every story page excludes them.
- 100 blank balances are treated as zero and flagged.
- Bands and segments are recomputed from Age, Tenure, Products and Balance (high balance = 100,000 or more).
- Customers with tenure longer than age minus 18, and ages of 80 or more, are kept and flagged.
- Credit scores above 850 occur only in the suspect rows.

Flagged records by block

	Rows 1-5,000 (reliable)	Rows 5,001-10,307 (suspect)
Customers	5,000	5,307
Churn rate	23.1%	51.3%
Blank balances	100	
Age 80 or over	470	1,349
Tenure > age - 18	570	403

Source: Bank_Churn.xlsx, all 10,307 records · 11

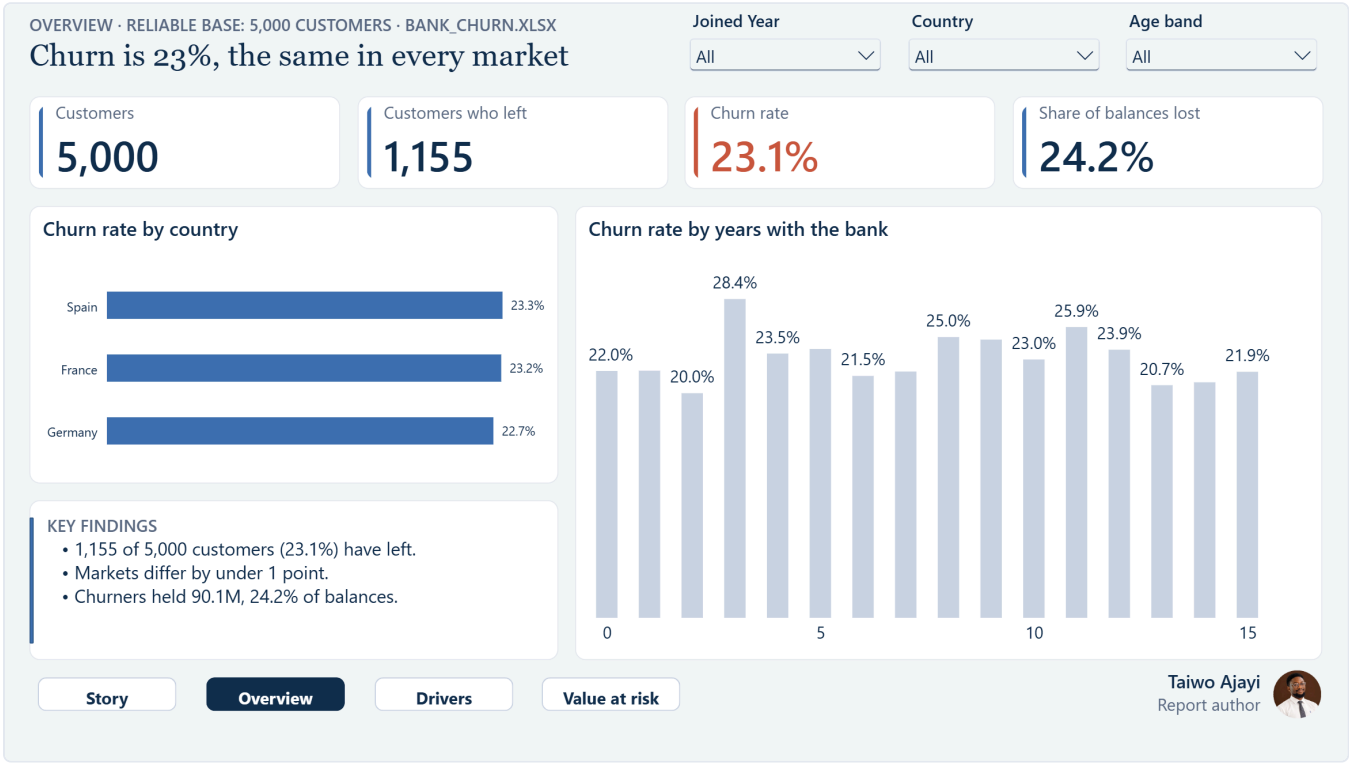
Dashboards

The assumptions behind the numbers, and a table of flagged records for each block. **This page is not filtered**, so both blocks appear.

Dashboards: three interactive pages for exploring the reliable base

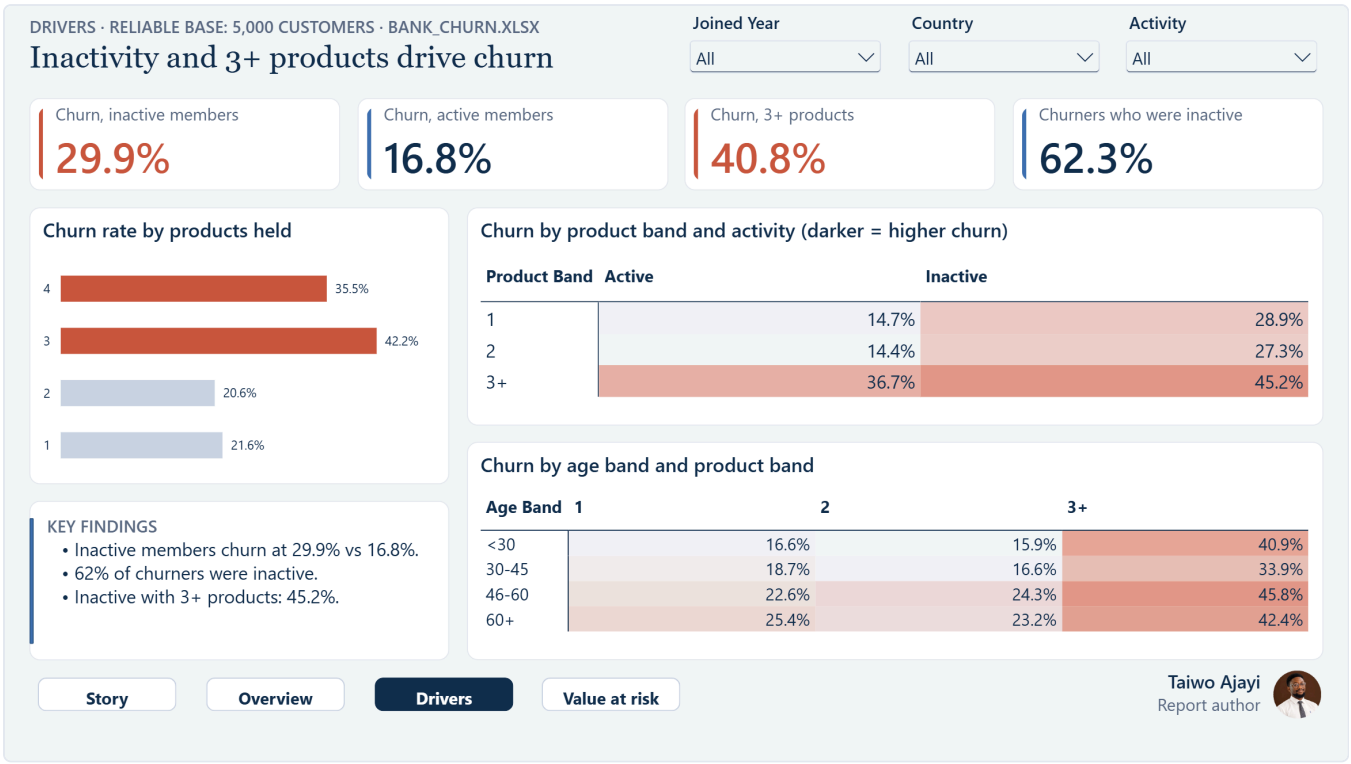
Each dashboard has a four-tile KPI row, Joined Year and Country slicers that stay in sync across all three, one page-specific slicer, tabs at the bottom for moving between them, and the report author's name and photo (Taiwo Ajayi) in the bottom-right corner. The **Key findings** tile describes the unfiltered reliable base and does not change with the slicers.

D1 · Churn overview



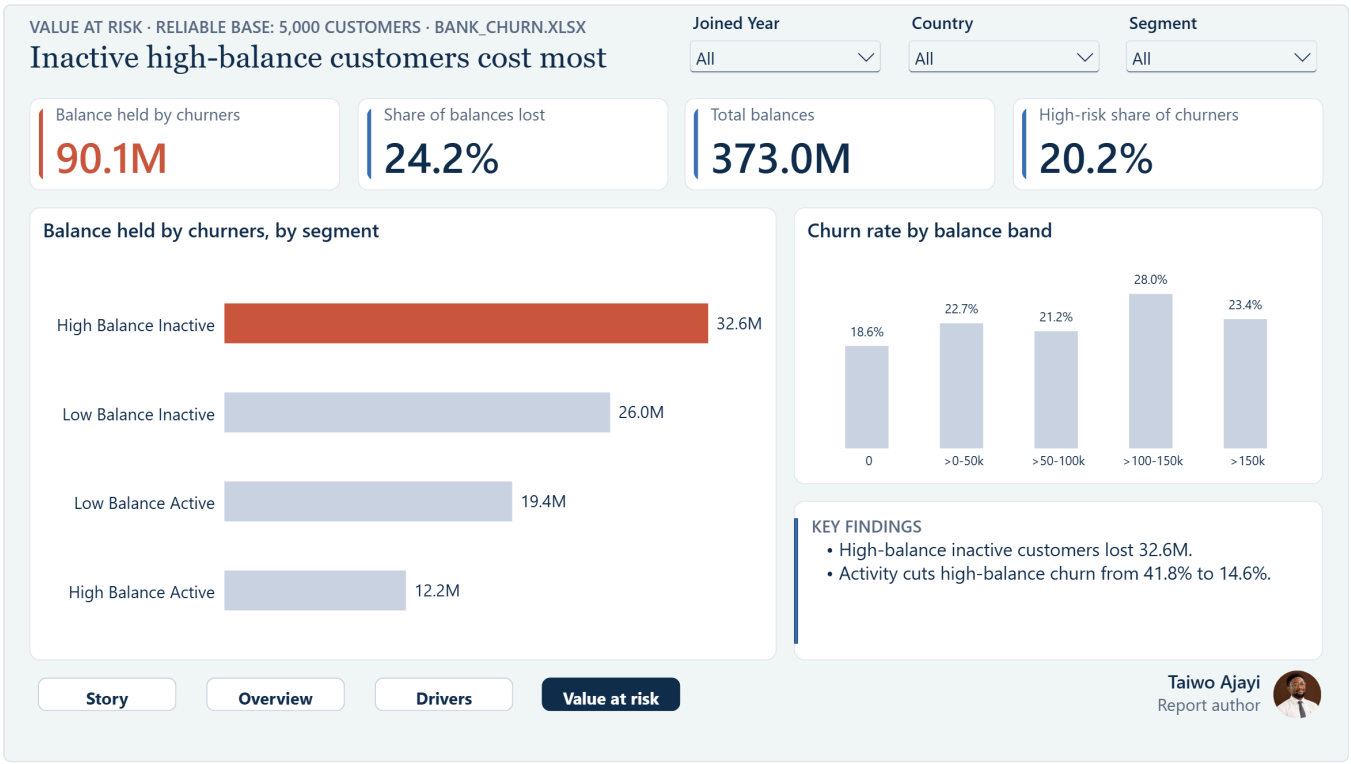
Slicers: Joined Year, Country, Age band. **KPIs:** customers, customers who left, churn rate, share of balances lost. **Charts:** churn by country (right-click a bar → Drill through → Customer list) and churn by years with the bank.

D2 · Customer drivers



Slicers: Joined Year, Country, Activity. **KPIs:** churn for inactive and active members, churn with 3+ products, share of churners who were inactive. **Charts:** churn by products held, and two gradient heat maps (product band × activity, age band × product band).

D3 · Value at risk



Slicers: Joined Year, Country, Segment. **KPIs:** balances held by churners, share of balances lost, total balances, high-risk share of churners. **Charts:** balance held by churners by segment (high-balance inactive highlighted) and churn by balance band.

Customer list (hidden drill-through)



Customer ID	Country	Age	Tenure (years)	Products	Activity	Balance	Churned
4832	Germany	21	4	2	Inactive	238,544	True
2103	Germany	74	12	1	Inactive	220,594	True
3007	France	85	0	2	Inactive	220,161	False
642	France	79	0	2	Active	210,231	False
4608	Spain	47	6	2	Active	209,556	False
2581	France	37	2	1	Inactive	203,607	False
794	Germany	64	14	2	Active	203,134	False
3016	France	49	8	2	Inactive	202,465	True
1521	France	79	3	1	Active	196,544	False
3355	Germany	80	14	1	Active	194,274	False
1136	Germany	81	0	2	Inactive	191,870	True
1464	France	54	9	2	Inactive	191,615	False
1306	France	35	10	1	Inactive	190,688	False
4785	Germany	68	11	1	Active	188,374	False
2808	Germany	75	6	1	Active	184,272	False
4283	Spain	73	6	2	Inactive	182,608	False
2248	Spain	38	5	1	Active	182,024	False

Opened from a country on D1. Lists every reliable customer in that market, highest balance first, with age, tenure, products, activity and churn status. **Back to dashboard** returns to the page you came from.

Reading, filtering, drilling and presenting

Navigating

- **Story:** start on the cover and click **Start the story**, then use the page tabs (or Page Down in presentation mode) for pages 2 to 11. Every story page has a **Dashboards** button in the footer.
- **Dashboards:** use the tabs at the bottom: **Story** (back to page 2), **Overview**, **Drivers**, **Value at risk**. The current page's tab is navy.
- In Power BI Desktop, hold **Ctrl** while clicking a button; in the Power BI service or reading view, a plain click works.

Filtering

- **Joined Year** and **Country** are synced: a choice on one dashboard applies on all three.
- The third slicer (Age band, Activity or Segment) applies only to its own page.
- Clicking a bar or cell cross-filters the other visuals on that page; click it again to clear.
- The reliable-record filter is a hidden page filter. Readers cannot remove it by accident.

Drilling through to customers

1. On D1, right-click a country bar in **Churn rate by country**.
2. Choose **Drill through → Customer list**.
3. Review the customers, then click **Back to dashboard**.

Presenting

- For a meeting, present pages 1 to 11 in order, then switch to the dashboards for questions.
- To share as a PDF: File → Export → Export to PDF. Story pages export as designed; dashboards export with their current slicer state.

Talk track

Page	One line to say
2	Our assumptions about safe customers are mostly wrong.
3	Half the file can't be trusted, so every number uses the reliable 5,000.
4	Inactive members nearly double churn and make up most of it.
5	A third product is a warning sign, not a loyalty sign.
6	Older customers leave a little more; tenure doesn't protect.
7	Inactivity and 3+ products stack up to 45%.
8	Our richest inactive customers are the costliest leavers.
9	Who the customer is doesn't explain churn; what they do does.
10	Two retention actions could cut churn to about 20%.

Calm, editorial and consistent: one accent colour per point

The report uses a restrained "corporate cool × editorial" look: a cool grey page, white tiles with hairline borders, Georgia serif titles and Segoe UI for everything else.

Navy #112D4E
Titles, text, active tabs

Blue #3F72AF
Neutral highlight, accent rules

Orange-red #C8553D
Problems only

Context #C9D3E3 / #DBE2EF
Non-highlighted bars

Page #F3F5F8
Canvas

Tile #FFFFFF
Visual containers

Hairline #E3E8EF
Tile borders

Heat gradient
#F3F5F8 → #E39A87

Typography

- Page titles: Georgia, 18-22 pt, written as a full-sentence finding
- Dashboard eyebrows: Segoe UI Semibold 12 pt, uppercase, muted
- KPI values: Segoe UI Semibold 28-30 pt
- Body, labels, axes: Segoe UI 12-14 pt; nothing below 12 pt

Layout

- Canvas 1280 × 720; 12-column grid, 24 px margins, 16 px gutters
- Title band on top, slicers top-right on dashboards
- KPI tiles with a 4 px left accent rule
- Footer row: source note and navigation

Accessibility

- Every visual has alt text describing the finding, not just the chart type.
- Colour is never the only signal: verdicts carry words, highlights are named in titles and notes.
- Blue and orange-red avoid the red/green pairing; navy text on white passes WCAG AAA.

Tie-out values for the reliable base (5,000 customers)

After any change to the data or model, these values should still appear in the report.

Check	Expected value
Customers / churned / churn rate	5,000 / 1,155 / 23.1%
Total balance / balance lost / share lost	373.0M / 90.1M / 24.2%
Churn: active / inactive	16.8% / 29.9%
Inactive share of customers / of churners	48.2% / 62.3%
Churn by products 1 / 2 / 3 / 4	21.6% / 20.6% / 42.2% / 35.5%
Churn 3+ products; share of customers / churn	40.8%; 10.1% / 17.7%
Churn by age band <30 / 30-45 / 46-60 / 60+	18.7% / 19.4% / 25.4% / 26.3%
Churn by country France / Germany / Spain	23.2% / 22.7% / 23.3%
Segments HBA / HBI / LBA / LBI	14.6% / 41.8% / 17.6% / 25.7%
High balance inactive: customers / balance / leavers / lost	624 / 77.6M / 261 / 32.6M
Active members with 1-2 products	14.5%
Scenario: prevented / churn after actions	172 (117.5 + 54.1) / 19.7%
Suspect block: rows / churn	5,307 / 51.3%
Data flags reliable: blank balance / age 80+ / tenure flag	100 / 470 / 570

If a figure differs, check first that IsReliable is TRUE on the page, that the three yes/no columns are true/false (not text), that blank balances are 0, and that the band keys are the rebuilt ones.

How to refresh, change and rebuild the report

Refreshing with new data

1. Replace `Bank_Churn.xlsx` with a file that has the same sheet and column names.
2. If the file moves, update the `SourceFile` parameter (Transform data → Manage parameters).
3. If the snapshot year changes, update `SnapshotYear`; the date table and join dates follow.
4. Click **Refresh**, then check the tie-out values in section 12.

Changing the report pages

The report pages are not edited by hand. They are generated by a Node.js script, so the whole report stays consistent:

```
cd "C:\Users\USER\Desktop\ALL PROJECTS\bannk Churn"
node _build\generate-report.js                      # rebuild theme + all 15 pages
powerbi-report-author validate "Bank Churn.Report"  # check the PBIR files
```

Then, in Power BI Desktop, click **Apply external changes** on the yellow banner (or close without saving and reopen the .pbip).

- Page content, titles, colours and layout are defined in `_build\generate-report.js`. Each page is one block, and the shared helpers (`chart`, `card`, `matrix`, `textbox`, `button`) set the design system.
- The script rebuilds every page from scratch. Edits made by hand in Desktop will be overwritten the next time it runs; make changes in the script instead.
- The approved design contract is in `_brief\report-spec.md`. Update it when the layout changes.

Changing the model

- Add or change measures in Power BI Desktop (Modeling tab) or through the Power BI modeling MCP, then press **Ctrl+S** so the TMDL files in `Bank Churn.SemanticModel` are updated.
- Keep measure names stable: the report pages reference them by name.

Known Power BI Desktop behaviours

Behaviour	What to do
Saving removes the `$schema` line from the .pbip and definition.pbir	Harmless for Desktop. The generator puts it back automatically.
After a reload, Desktop marks the report as having unsaved changes	Press Ctrl+S (then regenerate) or use Apply external changes before the next reload.
The preview tool cannot find the Microsoft Store version of Desktop	Open the .pbip by double-clicking it, or set `PBI_DESKTOP_PATH` to the Store install path.
Auto date/time creates hidden date tables	Turn it off: File → Options → Current file → Data load → Auto date/time.

What the numbers can and cannot tell you

- **One snapshot, no dates.** There is no trend, growth or seasonality. "Joined Year" is derived as 2025 minus tenure, so it is a cohort view, not a timeline.
- **Reliable base only.** Findings describe the 5,000 reliable records. If the suspect block turns out to be genuine, every rate would need re-reading.
- **Currency not stated.** Balances are unitless and shown in millions.
- **Correlation, not causation.** Inactivity and 3+ products go with higher churn; the report does not prove they cause it. Action 2 starts with an audit for that reason.
- **Scenario estimates are simple.** They assume each target is met and add the two actions together, which double-counts the overlap.
- **Implausible values kept.** Ages of 80+ and tenures longer than working life are flagged, not removed, because removing them would change the reliable base the deck was built on.
- **Static findings text.** The Key findings tiles and story-page notes are written text, so they don't change with slicers.

APPENDIX A

Project files

Path (in ALL PROJECTS\bannk Churn)	What it is
`Bank Churn.pbip`	Open this in Power BI Desktop
`Bank Churn.SemanticModel\`	The data model as TMDL text files (tables, measures, relationships)
`Bank Churn.Report\`	The report as PBIR files (pages, visuals, theme). Generated; do not edit by hand
`Bank_Churn.xlsx`	Source data
`Bank Churn Who Leaves and Why.pptx`	Executive deck the story pages follow
`Bank_Churn_PowerBI_Data_Model_Guide.md`, `…Build_Guide.md`	Original briefs (written for all 10,307 rows; superseded where they conflict with this guide)
`Bank_Churn_PowerBI_Theme.json`	Original colour theme, merged into the report theme
`_brief\report-spec.md`	Approved report spec and canonical design contract
`_build\generate-report.js`	Generator for the theme and all report pages
`_build\update-spec.js`	Helper used to update the design contract
`Bank_Churn_Report_Guide.pdf`	This guide

APPENDIX B

Build history and decisions

Step	Decision	Why
Data profiling	Use the 5,000 reliable rows; keep the rest flagged	Every quality signal breaks at row 5,000; the deck agrees, the old guides did not
Segment rule	High balance = 100,000 or more	Matches the executive deck
Keys	Rebuild all four band and segment keys	The supplied keys contradict the raw values
Measures	All on FactCustomer in display folders	Single fact table; avoids an empty measures table
Targets	Deck targets (inactive 25%, 3+ products 30%) instead of the old 32% target	The 32% target was set for the 37.6% figure from the unreliable full file
Page plan	11 story pages from the deck + 3 dashboards + drill-through	Approved spec
Fix round 1	Taller buttons, number formats in millions, no inner card boxes, no repeated subtitles, fuller layouts on pages 3, 7, 10 and 11	Clipped labels and formatting issues found in the PDF review
Dashboards v2	Bento grid of varied tiles	User preference
Dashboards v3	Corporate cool × editorial: grey page, white tiles, Georgia titles, KPI row with accent rules, gradient heat maps, quiet tabs	User asked for a more elegant and professional look
Story pages v3	Same design language applied to the story pages	Consistency across the whole report